

Software Engineering (Sect)

- 7 In a chemical reaction a compound X is formed from a compound Y . The masses in grams of X and Y present at time t seconds after the start of the reaction are x and y respectively. The sum of the two masses is equal to 100 grams throughout the reaction. At any time, the rate of formation of X is proportional to the mass of Y at that time. When $t = 0$, $x = 5$ and $\frac{dx}{dt} = 1.9$.

(i) Show that x satisfies the differential equation

$$\frac{dx}{dt} = 0.02(100 - x). \quad [2]$$

(ii) Solve this differential equation, obtaining an expression for x in terms of t . [6]

(iii) State what happens to the value of x as t becomes very large. [1]

i) $x + y = 100$

$$y = 100 - x$$

$$\frac{dx}{dt} = k(y)$$

$$\frac{dx}{dt} \quad (\because) x = 5, t = 0,$$

$$\frac{dx}{dt} = k(100 - x) \quad \frac{dy}{dx} = 1.9$$

$$1.9 = k(100 - 5)$$

$$k = 0.02$$

$$\frac{dx}{dt} = 0.02(100 - x)$$

$$(ii) \quad \frac{dx}{100 - x} = 0.02 dt$$

$$-\ln |100 - x| = 0.02t + c$$

$$100 - x = e^{-(0.02t + c)} \quad (\because) t = 0, x = 5$$

$$100 - 5 = e^{-c}$$

$$95 = e^{-c}$$

$$-c = \ln 95; \quad c = -\ln 95$$

$$100 - x = e^{\ln 95 - 0.02t}$$

$$100 - x = 95e^{-0.02t}$$

$$x = 100 - 95e^{-0.02t}$$

$$iii) \quad x = 100 - \frac{95}{e^{0.02 \cdot \infty}}$$

$$x = 100 - \frac{95}{\infty}$$

$$x = 100 - 0$$

$$x = 100$$

$\therefore x$ approaches 100 as t becomes very Large

$$(x-2)(\sqrt{x}+\sqrt{2})$$

$$\sqrt{2}+\sqrt{x}$$

Assignment Calculus

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$$\lim_{x \rightarrow 5} \frac{5}{x-5}$$

$$\frac{5}{0-5} = -1$$

$$\lim_{x \rightarrow 5} = -1$$

$$\lim_{x \rightarrow 0} \frac{\sqrt{x+3} - \sqrt{3}}{x}$$

$$\frac{\sqrt{x+3} - \sqrt{3}}{x} \cdot \frac{\sqrt{x+3} + \sqrt{3}}{\sqrt{x+3} + \sqrt{3}}$$

$$\frac{x+3-3}{x(\sqrt{x+3} + \sqrt{3})} = \frac{1}{x(\sqrt{x+3} + \sqrt{3})}$$

$$\lim_{x \rightarrow 0} \frac{1}{x(\sqrt{x+3} + \sqrt{3})} = \frac{1}{2\sqrt{3}}$$

$$\lim_{p \rightarrow 0} \left(1 + \frac{p}{5} \right)^{5/p}$$

$$\lim_{p \rightarrow 0} \left(1 + \frac{p}{5} \right)^{5/p} = e$$

$$\lim_{x \rightarrow 0} \frac{1 - \cos^3 x}{\sin^2 x}$$

$$\frac{(1 - \cos x)(1 + \cos^2 x + \cos x)}{(1 - \cos x)(1 + \cos x)}$$

$$\lim_{x \rightarrow 0} \frac{1 + \cos^2(0) + \cos(0)}{1 + \cos(0)} = \frac{3}{2}$$

$$\lim_{x \rightarrow 0} \frac{(1+3x)^{2/x}}{(1+3x)^{1/3x}} = e^6$$

$$(2) \lim_{x \rightarrow 7} \frac{x^2 - 49}{x - 7}$$

$$\lim_{x \rightarrow 7} \frac{(x-7)(x+7)}{x-7} = x+7$$

$$\lim_{x \rightarrow 7} x+7 = 7+7 = 14$$

$$(4) \lim_{x \rightarrow -1} \frac{x^3 + x^2}{x^2 - 1}$$

$$\frac{x^2(x+1)}{(x-1)(x+1)} = \frac{x^2}{x-1}$$

$$\lim_{x \rightarrow -1} \frac{(-1)^2}{-1-1} = -0.5$$

$$(6) \lim_{x \rightarrow \infty} x \sin(1/x) \quad y = \frac{1}{x}$$

$$\lim_{y \rightarrow 0} \frac{1}{y} \sin y = 1$$

$$(8) \lim_{x \rightarrow \infty} \sqrt{x^2 + x + 1} - x$$

$$\frac{(\sqrt{x^2 + x + 1} - x)(\sqrt{x^2 + x + 1} + x)}{\sqrt{x^2 + x + 1} - x}$$

$$\frac{1 + 1/x}{\sqrt{1 + 1/x + 1/x^2} + 1}$$

$$\lim_{x \rightarrow \infty} = \frac{1}{2}$$

$$(10) \lim_{x \rightarrow 2} \frac{\sqrt{x} - \sqrt{2}}{x - 2}$$

$$\frac{(\sqrt{x} - \sqrt{2})(\sqrt{x} + \sqrt{2})}{(x - 2)(\sqrt{x} + \sqrt{2})} = \frac{1}{\sqrt{2} + \sqrt{x}}$$

$$\lim_{x \rightarrow 2} \frac{1}{\sqrt{2} + \sqrt{x}} = \frac{1}{2\sqrt{2}}$$

$$1) \lim_{x \rightarrow 0} \frac{6^x - 3^x - 2^x + 1}{x^2}$$

$$\frac{(3^x \cdot 2^x) - 3^x - 2^x + 1}{x^2}$$

$$\frac{(3^x - 1) - (2^x - 1)}{x}$$

$$\lim_{x \rightarrow 0} \ln 2 \cdot \ln 3$$

$$13) \lim_{x \rightarrow 0} \frac{1 - \cos 2x}{x^2}$$

$$\frac{2 \sin^2 x}{x^2}$$

$$2 \times \frac{1}{1} = 2$$

$$(12) \lim_{x \rightarrow 0} \frac{e^{-2x} - e^{-11x}}{x}$$

$$\frac{e^{-2x} - 1}{x} - \frac{e^{-11x} - 1}{x}$$

$$-2 \left(\frac{e^{-2x} - 1}{2x} \right) - (-11) \left(\frac{e^{-11x} - 1}{-11x} \right)$$

$$-2(1) - (-11)(1) = 9$$

$$14) \lim_{x \rightarrow 0} \frac{x}{\tan x}$$

$$\frac{x}{\sin x / \cos x} = \frac{\cos x}{\sin x}$$

$$\lim_{x \rightarrow 0} = \frac{1}{1} = 1$$

L'Hospital Theorem (0/0 form)

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$$\lim_{x \rightarrow 0} \frac{\sin x}{x}$$

$$= \frac{\cos x}{1} = \cos x = \cos 0 = 1$$

$$(2) \lim_{x \rightarrow 0} \frac{x - \sin x}{x^3}$$

$$= \frac{1 - \cos x}{3x^2} = \frac{-(-\sin x)}{6x} = \frac{\sin x}{6x} = \frac{1}{6}$$

$$\lim_{x \rightarrow 0} \frac{e^x - x - 1}{xe^x - x}$$

$$\frac{e^x - 1}{2e^x + e^x - 1} = \frac{e^x}{3e^x} = \frac{1}{3}$$

$$\frac{e^x - 1}{2e^x + e^x - 1} = \frac{e^x}{3e^x} = \frac{1}{3}$$

$$(4) \lim_{x \rightarrow 0} \frac{\ln(\cos x)}{\ln(\cos 3x)}$$

$$\frac{-\sin x}{\cos x}$$

$$= \frac{-\sin x}{\cos x} = -\tan x = -\frac{\sec^2 x}{3 \sec^2 3x}$$

$$= \frac{3 \sin 3x}{\cos 3x}$$

$$= \frac{\cos^2 3x}{9 \cos^2 x} = \frac{1}{9}$$

$$\lim_{x \rightarrow 0} \frac{2e^x - 2\cos x}{x \sin x}$$

$$\frac{2e^x + 2\sin x}{x \cos x + \sin x} = \frac{2e^x + 2\cos x}{2\cos x - x \sin x}$$

$$= \frac{2 + 2}{2 - 0} = 2$$

$$(6) \lim_{x \rightarrow 0} \frac{\ln(1-x^2)}{\ln(\cos x)}$$

$$\frac{2x}{1-x^2} \times \frac{\cos x}{-\sin x} = \frac{-2x}{(1-x^2)\tan x}$$

$$= \frac{-2}{\sec^2 x (1-x^2) + \tan x (2x)} = -2$$

$$\lim_{x \rightarrow 0} \frac{\ln x^2}{\cot x^2}$$

$$\frac{2x}{x}$$

$$= -1$$

$$= \frac{-2x \operatorname{cosec}^2(x^2)}{x^2 \operatorname{cosec}^2(x^2)}$$

$$= \frac{-\sin^2(x^2)}{x^2} = -1$$

L'Hospital (0 · ∞)

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$$1) \lim_{x \rightarrow 0} x \ln(\tan x)$$

$$\frac{\ln(\tan x)}{1/x} = \frac{\sec^2 x}{\tan x} = -x^{-2}$$

$$-x^2 \sec^2 x$$

$$+ \tan x$$

$$= - (2x \sec^2 x + 2x^2 \sec^2 x \tan x)$$

$$\sec^2 x$$

$$- (2x + 2x^2 \tan x)$$

$$= 0$$

$$(3) \lim_{x \rightarrow \pi/2} (1 - \sin x) \tan x$$

$$\frac{1 - \sin x}{\cot x} = \frac{-\cos x}{-\csc^2(x)}$$

$$\frac{+\sin x}{-2x \sec^2(x) \cot x} = \frac{1}{2} \frac{\sin^4(x)}{1}$$

$$\frac{0}{2} = 0$$

$$(2) \lim_{x \rightarrow 0} \sin x \cdot \ln x$$

$$\frac{\sin x}{1/\ln x} = \frac{\cos x}{x} = -\sin x$$

$$(4) \lim_{x \rightarrow 0} x \tan\left(\frac{\pi}{2} - x\right)$$

$$x \cot(x)$$

$$\frac{x}{\tan x} = \frac{1}{\sec^2 x}$$

$$\cos^2 x = 1$$

L'Hospital (0^0 , ∞^0 , 1^∞)

$$(1) \lim_{x \rightarrow 0} (1+x)^{1/x}$$

$$= \exp \left(\frac{\ln(1+x)}{x} \right)$$

$$= \exp \left(\frac{1}{1+x} \right)$$

$$\exp(1) = e$$

$$(2) \lim_{x \rightarrow 5} \frac{(x-5)^{\cos x}}{(x-5)^{\sin x}}$$

$$\exp((x-5) \ln(x-5))$$

$$\exp \left(\frac{\ln(x-5)}{(x-5)^{-1}} \right)$$

$$\exp \left(\frac{1}{(x-5)} \cdot -(x-5)^2 \right)$$

$$\exp(5-x)$$

$$\exp(0) = 1$$

$$(3) \lim_{x \rightarrow 1} (1-x^2)^{1/\ln(1-x)}$$

$$\exp \left(\frac{\ln(1-x^2)}{\ln(1-x)} \right)$$

$$\exp \left(\frac{-2x/1-x^2}{-1/1-x} \right)$$

$$\exp \left(\frac{2x(1-x)}{(1-x)(1+x)} \right)$$

$$\exp \left(\frac{2x}{1+x} \right) = \exp \left(\frac{2}{2} \right)$$

$$= e$$

$$(2) \lim_{x \rightarrow 0} \frac{\ln(\tan x)}{x}$$

$$\exp \left(\frac{\ln(1/x) \cdot \tan x}{x} \right)$$

$$= \exp \left(\frac{\ln(1/x) \cdot \tan x}{x} \right)$$

$$\exp \left(\frac{\ln(1/x) \cdot \tan x}{x} \right) = \exp \left(\frac{\ln(1/x) \cdot \tan x}{x} \right)$$

$$\exp(1) = e$$

$$\exp \left(\frac{\ln(1/x)}{\cot x} \right)$$

$$\exp \left(\frac{1/x}{-\csc^2 x} \right)$$

$$= \exp \left(\frac{-\sin^2 x}{x} \right)$$

$$= \exp(-\sin^2 x)$$

$$\exp 0 = 1$$

$$(4) \lim_{x \rightarrow 0} (1 + \sin x)^{\cot x}$$

$$\exp \left(\frac{\ln(1 + \sin x)}{\tan x} \right)$$

$$\exp \left(\frac{\cos x / 1 + \sin x}{1/\cos^2 x} \right)$$

$$\exp \left(\frac{\cos^3 x}{1 + \sin x} \right) = \exp \left(\frac{1}{1} \right)$$

$$= e$$

$$(6) \lim_{x \rightarrow \pi/2} (1 - \sin x)^{\cos x}$$

$$\exp \left(\frac{\ln(1 - \sin x)}{\sec x} \right)$$

$$\exp \left(\frac{-\cos x}{(1 - \sin x)(\sec x \tan x)} \right)$$

$$\exp \left(\frac{-\cos^3 x}{\sin x (1 - \sin x)} \right)$$

$$\exp \left(\frac{-\cos^3 x \csc x}{1 - \sin x} \right)$$

$$\exp \left(-3\cos x + \cos^2 x \cot x \csc x \right)$$

$$\exp \left(-3\cos x + \frac{\cos^3 x}{\sin^2 x} \right)$$

$$\exp(0) = 1$$

$$(7) \lim_{x \rightarrow 0} (\cot x)^{\sin x}$$

$$\exp(\sin x \ln(\cot x))$$

$$\exp \left(\frac{-\csc^2 x}{(\cot x)(-\csc x \cot x)} \right)$$

$$\exp \left(\frac{\csc x}{\cot^2 x} \right) = \exp \left(\frac{\sin x}{\cos^2 x} \right)$$

$$\exp \left(\frac{0}{1} \right) = \exp(0) = 1$$

$$\lim_{x \rightarrow 0} \frac{1}{e^x - 1}$$

$$= \frac{x - e^x + x}{x e^x - x}$$

$$= \frac{-e^x}{2e^x + x e^x}$$

$$= \frac{-1}{2}$$

$$= \frac{-1}{2}$$

(*) ~~Indeterminate~~ $\infty - \infty$ forms :-

$$(1) \lim_{x \rightarrow 0} \left(\frac{1}{x} - \frac{1}{\sin x} \right)$$

$$\lim_{x \rightarrow 0} \left(\frac{\sin x - x}{x \sin x} \right)$$

$$= \frac{\cos x - 1}{\sin x + x \cos x} = \frac{-\sin x}{2\cos x - x \sin x}$$

$$= \frac{0}{2} = 0$$

$$(2) \lim_{x \rightarrow 0} \left(\frac{1}{x} - \cot x \right)$$

$$\frac{1}{x} - \frac{1}{\tan x} = \frac{\tan x - x}{x \tan x}$$

$$\frac{\sec^2 x - 1}{\tan x + x \sec^2 x}$$

$$= \frac{1}{\cos^2 x} - 1$$

$$\frac{\sin x}{\cos x} + \frac{x}{\cos^2 x}$$

$$= \frac{\sin^2 x}{\frac{1}{2} \sin 2x + x \cos 2x}$$

$$= \frac{0}{2} = 0$$

Maclaurin

Maclaurin

$f(x) = \sin x$

$f'(x) = \cos x$

$f''(x) = -\sin x$

$f'''(x) = -\cos x$

$f^{(4)}(x) = \sin x$

$$(\cot x)^{\sin x}$$

$$\ln(\cot x)$$

$$-\csc^2 x$$

$$\cot x)(-\csc x \cot x)$$

$$\csc x = \exp\left(\frac{\sin x}{\cos^2 x}\right)$$

$$\frac{0}{1} = \exp(0) = 1$$

$$\frac{1}{x} \cot x$$

$$1 = \tan x - x \tan x$$

$$-1$$

$$+ x \sec^2 x$$

$$1$$

$$+ \frac{x}{\cos^2 x}$$

$$2x + x$$

$$= 0$$

$$\sin 2x = \sin x \cdot -\sin^3 x + \sin^5 x$$

$$\frac{\sin^7 x}{7!} + \frac{\sin^9 x}{9!}$$

$$= 0$$

$$\lim_{x \rightarrow 0} \left(\frac{1}{e^x - 1} - \frac{1}{x} \right)$$

$$= \frac{x - e^x + 1}{xe^x - x} = \frac{1 - e^x}{e^x + xe^x - 1}$$

$$= \frac{-e^x}{2e^x + xe^x} = \frac{-1}{2 + x}$$

$$= \frac{-1}{2}$$

$$(4) \lim_{x \rightarrow 4} \left(\frac{1}{\ln(x-3)} - \frac{1}{x-4} \right)$$

$$= \frac{x-4 - \ln(x-3)}{(x-4)\ln(x-3)}$$

$$= 0 \cdot \frac{1}{x-3}$$

$$\frac{x-4}{x-3} + \frac{\ln(x-3)}{x-3}$$

$$= x-2$$

$$(x-4) + (x-3)\ln(x-3)$$

$$= 1$$

$$1 + \frac{x-4}{x-3} + \frac{\ln(x-3)}{x-3}$$

$$= \frac{1}{1} = 1$$

Maclaurins And Taylors Expansion:

find Maclaurin's Series of $f(x) = \sin x$

$$f(x) = \sin x \quad f(0) = 0$$

$$f'(x) = \cos x \quad f'(0) = 1$$

$$f''(x) = -\sin x \quad f''(0) = 0$$

$$f'''(x) = -\cos x \quad f'''(0) = -1$$

$$f^{(4)}(x) = \sin x \quad f^{(4)}(0) = 0$$

$$f(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} \dots$$

$$\sin 2x = \sin x \cdot -\sin^3 x + \sin^5 x$$

$$\frac{\sin^7 x}{7!} + \frac{\sin^9 x}{9!}$$

(a) find Maclaurin series of $f(x)$
 $f(x) = \ln(1-x)$

$$f(x) = \ln(1-x), \quad f(0) = 0$$

$$f'(x) = \frac{-1}{1-x}, \quad f'(0) = -1$$

$$f''(x) = (1-x)^{-2}, \quad f''(0) = 1$$

$$f'''(x) = -2(1-x)^{-3}, \quad f'''(0) = -2$$

$$f^{(4)}(x) = 6(1-x)^{-4}, \quad f^{(4)}(0) = 6$$

$$f(x) = -x + \frac{x^2}{2!} - \frac{2x^3}{3!} + \frac{6x^4}{4!} - \frac{24x^5}{5!} + \frac{120x^6}{6!} - \dots$$

(b) find Maclaurin series of

$$f(x) = e^x \cos x, \quad f(0) = 1$$

$$f'(x) = e^x \cos x - e^x \sin x, \quad f'(0) = 1$$

$$f''(x) = -2e^x \sin x, \quad f''(0) = 0$$

$$f'''(x) = -2e^x (\sin x + \cos x), \quad f'''(0) = -2$$

$$f^{(4)}(x) = -4e^x \sin x, \quad f^{(4)}(0) = 0$$

$$f(x) = 1 + x - \frac{2x^3}{3!} + \dots$$

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(c) find Maclaurin's Series of
 $f(x) = \cos^2 x$

$$f(x) = \cos^2 x, \quad f(0) = 1$$

$$f'(x) = -\sin 2x, \quad f'(0) = 0$$

$$f''(x) = -2\cos 2x, \quad f''(0) = -2$$

$$f'''(x) = 4\sin 2x, \quad f'''(0) = 0$$

$$f^{(4)}(x) = 8\cos 2x, \quad f^{(4)}(0) = 8$$

$$f^{(5)}(x) = -16\sin 2x, \quad f^{(5)}(0) = 0$$

$$f(x) = 1 - \frac{2x^2}{2!} + \frac{8x^4}{4!} - \dots$$

(d) find Maclaurin's Series of
 $f(x) = \ln(1+x)$

$$f(x) = \ln(1+x), \quad f(0) = 0$$

$$f'(x) = (1+x)^{-1}, \quad f'(0) = 1$$

$$f''(x) = -(1+x)^{-2}, \quad f''(0) = -1$$

$$f'''(x) = 2(1+x)^{-3}, \quad f'''(0) = 2$$

$$f^{(4)}(x) = -6(1+x)^{-4}, \quad f^{(4)}(0) = -6$$

$$f^{(5)}(x) = 24(1+x)^{-5}, \quad f^{(5)}(0) = 24$$

$$f(x) = x - \frac{x^2}{2!} + \frac{2x^3}{3!} - \frac{6x^4}{4!} + \frac{24x^5}{5!} - \dots$$

Series of

$$f(0) = 1$$

$$f'(0) = 0$$

$$f''(0) = -2$$

$$f'''(0) = 0$$

$$f^{(4)}(0) = 8$$

$$f^{(5)}(0) = 0$$

$$f^{(6)}(0) = 0$$

$$f^{(7)}(0) = 0$$

$$f^{(8)}(0) = 0$$

$$f^{(9)}(0) = 0$$

$$f^{(10)}(0) = 0$$

$$f^{(11)}(0) = 0$$

$$f^{(12)}(0) = 0$$

$$f^{(13)}(0) = 0$$

$$f^{(14)}(0) = 0$$

$$f^{(15)}(0) = 0$$

$$f^{(16)}(0) = 0$$

$$f^{(17)}(0) = 0$$

$$f^{(18)}(0) = 0$$

$$f^{(19)}(0) = 0$$

$$f^{(20)}(0) = 0$$

$$f^{(21)}(0) = 0$$

$$f^{(22)}(0) = 0$$

$$f^{(23)}(0) = 0$$

$$f^{(24)}(0) = 0$$

$$f^{(25)}(0) = 0$$

$$f^{(26)}(0) = 0$$

find Maclaurian's Series of

$$f(x) = \sinh(x)$$

$$f(x) = \sinh(x)$$

$$f'(x) = \cosh(x)$$

$$f''(x) = \sinh(x)$$

$$f'''(x) = \cosh(x)$$

$$f^{(4)}(x) = \sinh(x)$$

$$f(0) = 0$$

$$f'(0) = 1$$

$$f''(0) = 0$$

$$f'''(0) = 1$$

$$f^{(4)}(0) = 0$$

$$f(x) = x + \frac{x^3}{3!} + \frac{x^5}{5!} + \frac{x^7}{7!} + \dots$$

find Maclaurian's Series of

$$f(x) = \sin^{-1}(x)$$

$$f(x) = \sin^{-1}(x)$$

$$f'(x) = (1-x^2)^{-1/2}$$

$$f''(x) = \frac{x}{(1-x^2)^{3/2}}$$

$$f'''(x) = \frac{1+2x^2}{(1-x^2)^{5/2}}$$

$$f^{(4)}(x) = \frac{x(9+6x^2)}{(1-x^2)^{7/2}}$$

$$f^{(5)}(x) = \frac{9+72x^2+24x^4}{(1-x^2)^{9/2}}$$

$$f^{(6)}(x) = \frac{9+72x^2+24x^4}{(1-x^2)^{9/2}}$$

$$f^{(7)}(x) = \frac{9+72x^2+24x^4}{(1-x^2)^{9/2}}$$

$$f^{(8)}(x) = \frac{9+72x^2+24x^4}{(1-x^2)^{9/2}}$$

$$f^{(9)}(x) = \frac{9+72x^2+24x^4}{(1-x^2)^{9/2}}$$

$$f^{(10)}(x) = \frac{9+72x^2+24x^4}{(1-x^2)^{9/2}}$$

$$f^{(11)}(x) = \frac{9+72x^2+24x^4}{(1-x^2)^{9/2}}$$

$$f^{(12)}(x) = \frac{9+72x^2+24x^4}{(1-x^2)^{9/2}}$$

$$f^{(13)}(x) = \frac{9+72x^2+24x^4}{(1-x^2)^{9/2}}$$

$$f^{(14)}(x) = \frac{9+72x^2+24x^4}{(1-x^2)^{9/2}}$$

$$f^{(15)}(x) = \frac{9+72x^2+24x^4}{(1-x^2)^{9/2}}$$

$$f^{(16)}(x) = \frac{9+72x^2+24x^4}{(1-x^2)^{9/2}}$$

$$f^{(17)}(x) = \frac{9+72x^2+24x^4}{(1-x^2)^{9/2}}$$

$$f^{(18)}(x) = \frac{9+72x^2+24x^4}{(1-x^2)^{9/2}}$$

$$f^{(19)}(x) = \frac{9+72x^2+24x^4}{(1-x^2)^{9/2}}$$

$$f^{(20)}(x) = \frac{9+72x^2+24x^4}{(1-x^2)^{9/2}}$$

$$f^{(21)}(x) = \frac{9+72x^2+24x^4}{(1-x^2)^{9/2}}$$

$$f^{(22)}(x) = \frac{9+72x^2+24x^4}{(1-x^2)^{9/2}}$$

$$f^{(23)}(x) = \frac{9+72x^2+24x^4}{(1-x^2)^{9/2}}$$

$$f^{(24)}(x) = \frac{9+72x^2+24x^4}{(1-x^2)^{9/2}}$$

$$f^{(25)}(x) = \frac{9+72x^2+24x^4}{(1-x^2)^{9/2}}$$

$$f^{(26)}(x) = \frac{9+72x^2+24x^4}{(1-x^2)^{9/2}}$$

$$f^{(27)}(x) = \frac{9+72x^2+24x^4}{(1-x^2)^{9/2}}$$

$$f^{(28)}(x) = \frac{9+72x^2+24x^4}{(1-x^2)^{9/2}}$$

$$f^{(29)}(x) = \frac{9+72x^2+24x^4}{(1-x^2)^{9/2}}$$

$$f^{(30)}(x) = \frac{9+72x^2+24x^4}{(1-x^2)^{9/2}}$$

find Taylor expansion

$$f(x) = \ln x \text{ in } (x-1)$$

$$f(x) = \ln x$$

$$f'(x) = x^{-1}$$

$$f''(x) = -x^{-2}$$

$$f'''(x) = 2x^{-3}$$

$$f^{(4)}(x) = -6x^{-4}$$

$$f^{(5)}(x) = 24x^{-5}$$

$$f(a) = 0$$

$$f'(a) = 1$$

$$f''(a) = -1$$

$$f'''(a) = 2$$

$$f^{(4)}(a) = -6$$

$$f^{(5)}(a) = 24$$

$$f(x) = (x-1) - \frac{(x-1)^2}{2!} + \frac{2(x-1)^3}{3!} - \frac{6(x-1)^4}{4!} + \frac{24(x-1)^5}{5!} - \dots$$

(Q) find Taylor expansion of

$$2) e^{-x} \text{ in } (x+1)$$

$$f(x) = e^{-x}$$

$$f'(x) = -e^{-x}$$

$$f''(x) = e^{-x}$$

$$f'''(x) = -e^{-x}$$

$$f^{(4)}(x) = e^{-x}$$

$$f^{(5)}(x) = -e^{-x}$$

$$f^{(6)}(x) = e^{-x}$$

$$f^{(7)}(x) = -e^{-x}$$

$$f^{(8)}(x) = e^{-x}$$

$$f^{(9)}(x) = -e^{-x}$$

$$f^{(10)}(x) = e^{-x}$$

$$f^{(11)}(x) = -e^{-x}$$

$$f^{(12)}(x) = e^{-x}$$

$$f^{(13)}(x) = -e^{-x}$$

$$f^{(14)}(x) = e^{-x}$$

$$f^{(15)}(x) = -e^{-x}$$

$$f^{(16)}(x) = e^{-x}$$

$$f(-1) = e$$

$$f'(-1) = -e$$

$$f''(-1) = e$$

$$f'''(-1) = -e$$

$$f^{(4)}(-1) = e$$

$$f^{(5)}(-1) = -e$$

$$f^{(6)}(-1) = e$$

$$f^{(7)}(-1) = -e$$

$$f^{(8)}(-1) = e$$

$$f^{(9)}(-1) = -e$$

$$f^{(10)}(-1) = e$$

$$f^{(11)}(-1) = -e$$

$$f^{(12)}(-1) = e$$

$$f^{(13)}(-1) = -e$$

$$f^{(14)}(-1) = e$$

$$f^{(15)}(-1) = -e$$

$$f^{(16)}(-1) = e$$

$$f(x) = e - e(x+1) + \frac{e(x+1)^2}{2!} - \frac{e(x+1)^3}{3!} + \frac{e(x+1)^4}{4!} - \frac{e(x+1)^5}{5!} + \dots$$

(10) Find Taylor Expansion of
3) $\ln(\sin x)$ in $(x-2)$

$\Rightarrow$ for $x=0$

put $x=0.1$

$$f(0.1) = 5(0.1)^4 - 20(0.1)^3 + 15(0.1)^2 = 0.13$$

put $x=-0.1$

$$f'(-0.1) = 5(-0.1)^4 - 20(-0.1)^3 + 15(-0.1)^2 = 0.17$$

$$f(x) = \ln(\sin 2) + (\cot 2)(x-2) - \frac{1}{2}(\csc^2(2))(x-2)^2 + \frac{1}{6}2\csc^2(2)\cot(2)(x-2)^3 \dots$$

at point $(3, 8)$ that curve is minimum

8) find Extreme high and low points.

at point $(1, 0)$ the curve maximum.

$$1) f(x) = x^5 - 5x^4 + 5x^3 - 1$$

$$f'(x) = 5x^4 - 20x^3 + 15x^2$$

$$5x^4 - 20x^3 + 15x^2 = 0$$

$$x^2 = 0$$

$$5x^2 - 20x + 15 = 0$$

$$x = 0$$

$$x = 3, x = 1$$

$\Rightarrow$ for $x=3$

put $x=3.1$

$$f'(3.1) = 5(3.1)^4 - 20(3.1)^3 + 15(3.1)^2 = 10.1$$

put $x=2.9$

$$f'(2.9) = 5(2.9)^4 - 20(2.9)^3 + 15(2.9)^2 = -7.99$$

$\Rightarrow$ for $x=1$

put $x=1.1$

$$f'(1.1) = 5(1.1)^4 - 20(1.1)^3 + 15(1.1)^2 = -1.14$$

put $x=0.9$

$$f'(0.9) = 5(0.9)^4 - 20(0.9)^3 + 15(0.9)^2 = 0.85$$

$$2) x^3 + 6x^2 - 15x + 5 = f(x)$$

$$f'(x) = 3x^2 + 12x - 15$$

$$f'(x) = 0$$

$$3(x^2 + 4x - 5) = 0$$

$$(x-1)(x+5) = 0$$

$$x = 1$$

$$x = -5$$

$\Rightarrow x=1$

put $x=1.1$

$$f'(1.1) = 3(1.1)^2 + 12(1.1) - 15 = 1.1$$

put $x=0.9$

$$f'(0.9) = 3(0.9)^2 + 12(0.9) - 15 = -0.9$$

$\Rightarrow x=-5$

put $x=-5.1$

$$f'(-5.1) = 3(-5.1)^2 + 12(-5.1) - 15 = -1.1$$

put $x=-4.9$

$$f'(-4.9) = 3(-4.9)^2 + 12(-4.9) - 15 = 0.9$$

$$f(-5) = (-5)^3 + 6(-5)^2 - 15(-5) + 5$$

$$f(-5) = 105$$

$$f(1) = (1)^3 + 6(1)^2 - 15(1) + 5$$

$$f(1) = -3$$

(1, -3) Minimum point

(-5, 105) Maximum point

$$f(x) = x^5 - 5x^3 + 5x + 1$$

$$f'(x) = 5x^4 - 15x^2 + 5$$

$$f'(x) = 0$$

$$5x^4 - 15x^2 + 5 = 0$$

$$x = \pm \sqrt{\frac{3 \pm \sqrt{5}}{2}} \quad x = \pm \sqrt{\frac{3 - \sqrt{5}}{2}}$$

$$\text{for } x = \sqrt{3 + \sqrt{5}}/2 = 1.61$$

$$x = 1.71$$

$$f'(1.71) = 5(1.71)^4 - 15(1.71)^2 + 5 = 4.29$$

$$x = 1.51$$

$$f'(1.51) = 5(1.51)^4 - 15(1.51)^2 + 5 = -3.01$$

$$\text{for } x = -\sqrt{3 + \sqrt{5}}/2 = -1.61$$

$$x = -1.71$$

$$f'(-1.71) = 5(-1.71)^4 - 15(-1.71)^2 + 5 = 4.29$$

$$x = -1.51$$

$$f'(-1.51) = 5(-1.51)^4 - 15(-1.51)^2 + 5 = -3.01$$

$$\text{for } x = \sqrt{3 - \sqrt{5}}/2 = 0.62$$

$$f'(0.62) = 5(0.62)^4 - 15(0.62)^2 + 5 = 1.33$$

$$x = 0.72$$

$$f'(0.72) = 5(0.72)^4 - 15(0.72)^2 + 5 = -1.40$$

$$f'(0.52) = 5(0.52)^4 - 15(0.52)^2 + 5 = 1.33$$

Hence the curve is max at points

$$\left(\sqrt{\frac{3 - \sqrt{5}}{2}}, 3 \right),$$

$$\left(-\sqrt{\frac{3 - \sqrt{5}}{2}}, -1 \right)$$

The curve is minimum at

$$\left(\sqrt{\frac{3 + \sqrt{5}}{2}}, -1 \right)$$

$$\left(-\sqrt{\frac{3 + \sqrt{5}}{2}}, 3 \right)$$

$$4) f(x) = 3x^4 - 4x^3 + 5$$

$$f'(x) = 12x^3 - 12x^2$$

$$f'(x) = 0$$

$$12x^2(x - 1) = 0$$

$$x = 0$$

$$\text{for } x = 0$$

$$x = 0.1$$

$$f'(0.1) = 12(0.1)^3 - 12(0.1)^2 = -0.108$$

$$x = -0.1$$

$$f'(-0.1) = 12(-0.1)^3 - 12(-0.1)^2 = -0.13$$

$$\text{for } x = 1$$

$$f'(1.1) = 12(1.1)^3 - 12(1.1)^2 = 1.42$$

$$f'(0.9) = 12(0.9)^3 - 12(0.9)^2 = -0.97$$

The curve has only minimum point at (1, 4)